

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

M.Tech. (CE/IT) Semester - I University Examination, 2013

CE704: Soft Computing

Date: 10.05.2013, Friday Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION-I

Q-1 Answer the following questions.

- 1 What is soft computing? How it differs from hard computing? [03]
- 2 What is hybrid system? Explain types of hybrid system with example. [04]

Q-2

- [A] What is Fuzzy set? How it is differ from Crisp set? Explain with example. [04]
- [B] Explain term Fuzzy Relation. How it is differ from simple Relation? List various Fuzzy Relation Operations with example [05]
- [C] What is Mamdani Fuzzy Model? How it is used for Fuzzy inference system? Explain with example. [05]

OR

- [B] What is Fuzzy inference system? Draw block diagram of fuzzy inference system. Explain its components and functions. [05]
- [C] What is Sugeno Fuzzy Model? How it is used for Fuzzy inference system? Explain with example. [05]

Q-3

- [A] Describe terms: Membership Function, Support, Core, Normality, and Convexity. [05]
- [B] Define and explain with equation and diagram: various Fuzzy set operation. [04]
- [C] Let $X=\{p,q,r,s\}$ $Y=\{1,2,3,4\}$. $A=\{(p,0),(q,0.8),(r,0.6),(s,1)\}$ $B=\{(1,0.2),(2,1),(3,0.8),(4,0)\}$ $C=\{(1,0),(2,0.4),(3,1),(4,0.8)\}$ Determine the implication relations: IF x is A THEN y is B. [05]

OR

- [B] Define all types of MF Formulations for MFs with one dimension. [04]
- [C] Explain with example: Fuzzy Rule based System. [05]

SECTION-II

Q-4 Answer the following questions.

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| 1 | Define Schema theorem and conversions in GA. | [03] |
| 2 | Define an artificial neural network. List application area of it. | [04] |

Q-5

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|-----|---|------|
| [A] | What is crossover in GA? Explain uniform crossover with example. | [04] |
| [B] | What is mutation rate? Explain mutation with example. | [05] |
| [C] | What is hybrid system? Explain types of hybrid system with example. | [05] |

OR

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|-----|--|------|
| [B] | What is GA? Explain benefits of GA. Write Applications of GA. | [05] |
| [C] | Define encoding. Explain various types of encoding system in GA. | [05] |

Q-6

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|-----|---|------|
| [A] | What is Defuzzification? Explain various methods with its equation. | [05] |
| [B] | State the properties of the processing element of an artificial neural network. What is significance of activation function used in perceptron network? | [04] |
| [C] | What is Hopfield Network? Explain Hopfield with example of Traveling Salesperson problem. | [05] |

OR

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| [B] | Explain Backpropagation Multilayer Perceptrons with example. | [04] |
| [C] | What is perceptron? List the limitations of perceptron .What is the importance of threshold in perceptron network. | [05] |